

A novel cotton mapping index combining Sentinel-1 SAR and Sentinel-2 multispectral imagery

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ABSTRACT

Cotton is an important cash crop in the world, as the main source of natural and renewable fiber for textiles. Accurate and timely monitoring of the cotton distribution is crucial for cotton cultivation management and international trade. However, most of the previous researches on cotton identification using remotely sensed images are highly dependent on training samples, and the collection of samples is time-consuming and expensive. To overcome this limitation, a new index, termed as Cotton Mapping Index (CMI), was developed in this study for automatic cotton mapping using time series of Sentinel-1 synthetic aperture radar (SAR) and Sentinel-2 Multi-spectral Instrument (MSI) satellite data. Four sites in the United States (U.S.) and four sites in China were selected to develop and assess the performance of the CMI. The spectral characteristics derived from Sentinel-2 and backscattering coefficients derived from Sentinel-1 for cotton and non-cotton crops during the cotton growth period were analyzed. Considering the phenology differences of crops in different regions, the features at an adaptive window were adopted to construct the CMI. The results showed that at the peak greenness period, the multiplication of red-edge 1 and red-edge 2 band for cotton samples were much larger than those for non-cotton samples, whereas the spectral angle at the red band as well as the absolute values of backscattering coefficients in vertical transmit and vertical receive (VV) polarization for cotton samples were much smaller than those for non-cotton samples. Based on these findings, the CMI was developed to identify cotton cultivated area within the cropland area. The overall accuracy of classification results for the sites in the U.S. was higher than 81.20%, and the mean relative error for the sites in Xinjiang of China was 26.69%. The CMI, which incorporated optical and radar features, had a better performance than the indices using optical features solely. The advantage of the CMI over supervised classifiers (i.e., k-nearest neighbors, support vector machine and random forest) is that no training samples are required. Moreover, the cotton distribution map can be obtained before the harvest using the CMI. These results indicated the potential of the CMI for cotton mapping. The applicability of CMI in other regions with different cropping systems and crop types needs to be further assessed in the future study.

1. Introduction

Cotton is an important cash crop, which plays an significant role in economic development (Razzaq et al., 2021). As the main natural fiber, it accounts for 3 percent of the agricultural area and meets 27 percent of the textile demand in the world (ICAC, 2018). The cotton cultivated area information is of significance for a series of issues that influence human well-being, ranging from textiles in the clothing manufacturing industry, cosmetics, medical gauze, to international trade (Akter et al., 2019; H. Li et al., 2021). The timely and accurate acquisition of spatial distribution

information of cotton provides the basis for yield estimation, agricultural production management and decision making.

The traditional methods for the acquisition of crop cultivated area are based on field measurement and statistical sampling, which are time-consuming, low efficiency and labour-intensive (Zhang et al., 2014). In the past few decades, satellite remote sensing has contributed to crop type mapping on a regional or global scale because of its advantages such as wide coverage, high integration, fast speed and periodicity (Chamorro Martinez et al., 2021). According to the number of images used, remote sensing-based studies on cotton identification can be

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divided into two categories, including the identification based on single-temporal and multi-temporal images. Li et al. (2011) extracted the cotton cultivated area by combining the decision tree classifier with the spectral characteristics, which were derived from the Huanjing image acquired on 1st July in Dingzhuang of China. However, it is difficult to extend the classification rules and parameters on a large scale, as different crop types in different regions may exhibit similar spectral characteristics at the same time. To solve this problem, some researchers have used multi-temporal images that can reflect crop phenology information for cotton mapping. For instance, Musande et al. (2012) combined multi-temporal indices (i.e., normalized difference vegetation index (NDVI), Transformed Normalized Difference Vegetation Index (TNDVI), Simple Ratio (SR), Triangular Vegetation Index (TVI) and Soil-Adjusted Vegetation Index (SAVI)) derived from Advanced Wide Field Sensor (AWIFS) and Linear Imaging Self Scanner Sensor (LISS-III) images with fuzzy classification for cotton mapping in Aurangabad of India. The results showed that the flowering-boll opening and harvesting stages were the optimum temporal for cotton identification. Based on the annual time series of vegetation index derived from Moderate Resolution Imaging Spectroradiometer (MODIS) images, Zhong et al. (2011), Conrad et al. (2011) and Xun et al. (2021) combined the phenological metrics, temporal segmentation metrics and harmonic features with the decision tree classification, classification and regression tree (CART) as well as fused representation-based algorithm for cotton identification, respectively. Based on the Sentinel-2 images acquired from March to November, Huang et al. (2018) combined spectral bands (i.e., blue, green, red, near infrared, red-edge bands), spectral indices (i.e., NDVI and Modified Chlorophyll Absorption Ratio Index (MCARI)) with the random forest classifier to map the spatial distribution of cotton and other crops in Shawan county of China. The results showed that involving the red-edge spectral band and related index can improve the classification accuracy.

However, all the above-mentioned methods belong to the supervised classifiers, and their accuracy highly depends on the high quality and sufficient quantity of samples to train the classifier (Zhu et al., 2016). It's usually a time-consuming, expensive and labor-intensive task to collect field samples (Ashourloo et al., 2019; Zhang and Roy, 2017; Zhong et al., 2019). The index-based mapping method has the advantage of simplicity and efficiency, which has been used in the identification of water (L. Li et al., 2021), plastic greenhouse (Yang et al., 2017), plastic materials (Guo and Li, 2020), and urban biophysical compositions (Deng and Wu, 2012). For specific crop type mapping, some features/indices have also been developed, such as for alfalfa (Ashourloo et al., 2018), maize (Qiu et al., 2018), canola (Ashourloo et al., 2019), soybeans (Silva Junior et al., 2020), potato (Ashourloo et al., 2020), and winter wheat (Qu et al., 2021). According to the spectral characteristics of red and near infrared bands during the growth periods of alfalfa and potato, Ashourloo et al. (2018, 2020) proposed two novel features to identify the alfalfa and potato areas, respectively, and the overall accuracy was over 90%. Silva Junior et al. (2020) proposed the Perpendicular Crop Enhancement Index (PCEI) for soybean mapping in Mato Grosso of Brazil, and the high PCEI values indicated the high probability of soybean areas. In terms of cotton mapping, L. Wang et al. (2019) proposed the Cotton Extraction Index (CEI) based on a single temporal of RapidEye image, which is the summation of the reflectance of red-edge and near infrared bands. The result showed that the classification accuracy achieved by the CEI was similar to those by the maximum likelihood and random forest classifiers. However, the selection of the acquisition time of satellite image is also critical to calculate the CEI for cotton mapping, and the differences of crop phenology over large regions will also affect the classification accuracy using single temporal image.

The features/indices proposed by the above-mentioned studies are mainly based on optical images. Synthetic Aperture Radar (SAR) imagery has posed a key advantage over optical images in crop mapping due to its all-weather acquisition capability (Inglada et al., 2016; Teimouri et al., 2019). Recent studies have shown that combining optical

and radar images has a better performance than using single sensor in the fields of land cover mapping (Ienco et al., 2019), crop classification (Orynbaikyzy et al., 2019; Sonobe et al., 2017) and biophysical metrics estimation (J. Wang et al., 2019). Steinhäuser et al. (2018) combined Sentinel-1 and Sentinel-2 images to classify land use and land cover types in Chennai Basin, and the results showed that the classification accuracy obtained by the combined images was improved compared with that by Sentinel-2 images solely. Adrian et al. (2021) utilized the backscattering coefficients, texture and spectral features derived from the multi-temporal Sentinel-1 SAR and Sentinel-2 optical images for classification, and the results showed that the fused radar and optical features achieved a higher overall accuracy for crop type mapping than those using radar or optical data solely.

In this study, a new index, called the Cotton Mapping Index (CMI), was proposed to map the cotton cultivated area combining Sentinel-1 SAR and Sentinel-2 Multispectral Instrument (MSI) images. Based on the preliminary analysis of the optical and radar features of cotton and non-cotton samples during the cotton growth period, three assumptions were proposed to select the optimal features to distinguish cotton from non-cotton crops. Considering the phenology difference in various regions, an adaptive window was introduced to develop the CMI. The main objectives of the current study are (1) to develop a novel index (CMI) for automatic cotton mapping; (2) to evaluate the capability and robustness of the proposed CMI in different study sites; and (3) to assess the performance of the CMI by comparing with the supervised classifiers.

2. Materials and methods

The overall workflow is shown in Fig. 1. Firstly, time series of radar and optical features for cotton and non-cotton samples were derived from multi-temporal Sentinel-1 SAR and Sentinel-2 MSI satellite data. Second, three assumptions were proposed based on the analysis of the spectral reflectance, spectral angle and backscattering coefficients of cotton and non-cotton samples during the cotton growth period. The appropriate features were then selected to develop the CMI. Finally, the proposed CMI was used to map the cotton cultivated area within the cropland mask, and its performance was evaluated and compared with those of the supervised classifiers.

2.1. Study sites

The phenological period and growth status of each crop are varied in different regions due to the influence of crop varieties, environmental conditions and farmers' decisions (Pan et al., 2012). Considering the availability of field data, a total of eight study sites were selected to test the potential of the proposed index, including four sites from the United States (U.S.), as well as four sites from China. The distribution of these study sites is shown in Fig. 2.

In the U.S., cotton is mainly cultivated in the southern region. Texas, Georgia, Mississippi, Oklahoma and Arkansas are the top five states with the largest cotton plantations in 2019. Arkansas has a humid subtropical climate, with the average maximum temperature of 34.2 °C in summer, average minimum temperature of −3°C in winter, and annual precipitation of 1220 mm. In this study, site A is selected from the northeast of Arkansas State, where the land is generally flat and soil is fertile. The main crops in this site include corn, cotton, soybeans and rice. According to the Crop Progress and Condition graphical products published by the United States Department of Agriculture (USDA), cotton and soybeans are usually planted from April to June/ July, and harvested from September to November. Corn and rice are usually planted from April to May/ June, and harvested from August to October. In this study, site A was used to develop the index for cotton identification, and the other study sites were used as implementation sites to assess the potential and stability of the proposed index. Site B is selected from the eastern Arkansas and western Mississippi, with soybeans, cotton, corn and rice as the main crops. Mississippi has a humid subtropical climate with long,

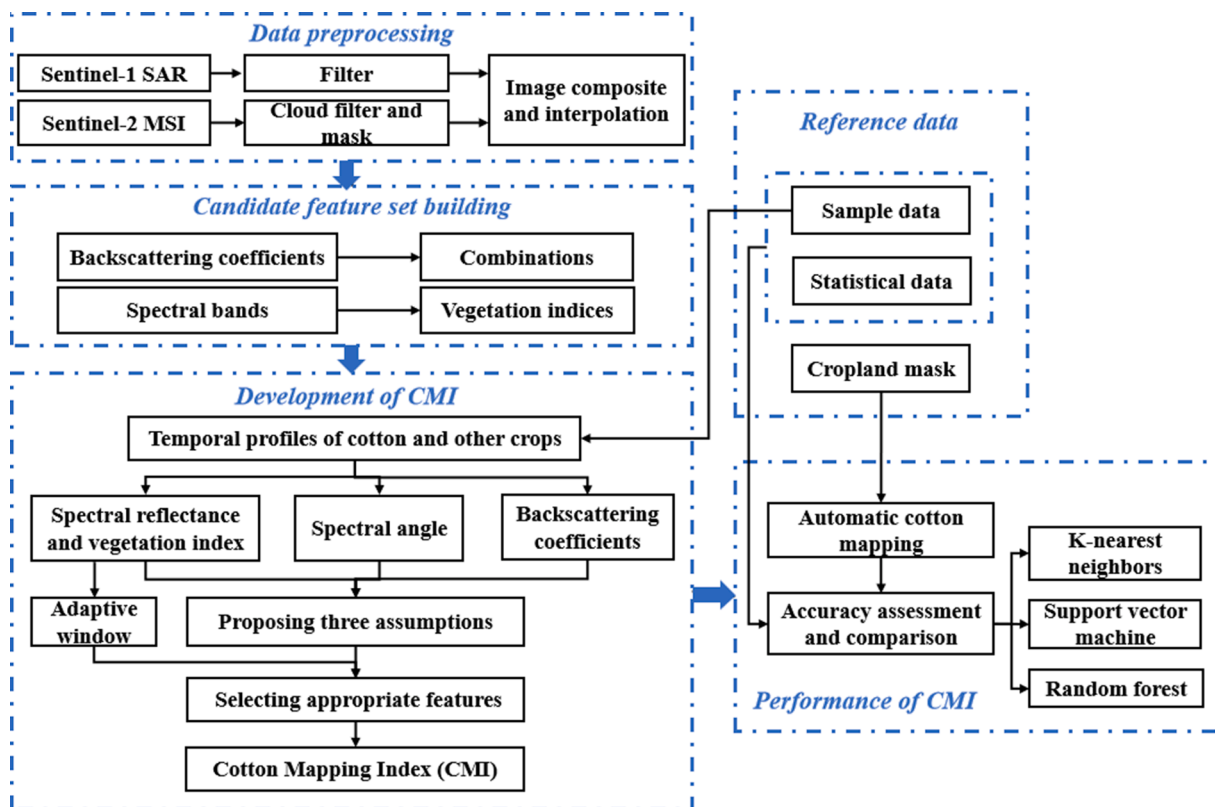


Fig. 1. Flowchart of the identification of cotton cultivated area in this study.

hot and humid summers, as well as short and mild winters. The average temperature is around 27 °C in July and 6 °C in January. It is entirely composed of lowlands, with an average elevation of 91 m. Georgia has a humid subtropical climate with hot and moist summers and mild winters. The daytime temperature in summer often exceeds 35 °C, and winter is characterized by mild temperature. The entire state receives moderate to heavy precipitation, varying from 1100 mm to 1900 mm. Site C is selected from the southern region of Georgia State, where the main crops include cotton, peanuts, corn and pecans. Site D is selected from the southern region of Texas State. It is located in the coastal plain, and the terrain is flat. This site has a humid subtropical climate. Summer is hot and humid, with the average daily temperature above 32.2 °C. The temperature reaches freezing point only a few times in winter. The main crops in this site include cotton, sorghum, corn and wheat.

Located in northwest of China, Xinjiang Uygur Autonomous Region is the largest cotton production region in China, accounting for 84.94 percent of the national total production in 2019 (National Bureau of Statistics of China, 2020). Xinjiang is in the arid and semiarid inland and has a typical temperate continental climate characterized by strong evaporation and rare rainfall, with a mean annual precipitation of 100–200 mm. The average temperature in January and July is −17.1 °C and 27.1 °C, respectively (Wang et al., 2012). To facilitate accuracy assessment using statistical data, ten counties from Xinjiang were selected and grouped as the four study sites (E–H), including Awat, Jiashi, Kuqa, Luntai, Makit, Jinghe, Wensu, Bole, Shule and Yopurga county (Fig. 2). These sites are dominated by single cropping system, and the main crops include cotton, wheat, maize, watermelon, grape and so on. The growing season of cotton is usually from April to October.

2.2. Data

2.2.1. Sentinel-2 images and preprocessing

The Sentinel-2 Multi-Spectral Instrument (MSI) onboard the Sentinel-2A and –2B satellites is developed by the European Space

Agency (ESA), which can capture images with a spatial resolution of 10–60 m and a revisit period of 5 days. It provides 13 spectral bands ranging from visible to shortwave infrared spectrum. The Level-2A surface reflectance (SR) images were used for cotton identification at sites A–D in this study, which have been available from the data catalogue of the Google Earth Engine (GEE) since 2019. Among all spectral bands, the 9 bands were used as the optical candidate features for cotton identification in this study, as shown in Table 1. In addition, the QA60 band, a bitmask band with cloud mask information, was used to mask opaque and cirrus clouds. Bands QA60, B5, B6, B7, B11 and B12 were resampled from 60 m/20 m to 10 m using the nearest neighbor resampling method in the GEE platform.

Taking the sites A–D as an example, the number of all available images with different cloud percentages is shown in Fig. 3. For each region, there were over 50 percent images with the cloud cover less than 50%. In this study, all available Sentinel-2 observations covering the sites A–C during April–November in 2019 with cloud coverage of less than 50% were considered. For site D, due to the limited number of high-quality images available, the images in 2020 were used. In total, 91, 168, 68 and 255 Sentinel-2 images for sites A–D composed the initial image data for cotton mapping. The Level-1C top-of-atmosphere (TOA) reflectance images for sites E–H in 2018 were used due to the unavailability of statistical data in 2019 when we conducted the experiment.

There was more than one satellite track involved to achieve full coverage for each study site. The number of images varies for each site, and more observations can be found along the strips where satellite orbits overlap each other. In some regions, there is more than one observation at different times on the same day, which may lead to differences in the observed values of spectral bands. In addition, the image selection concerning cloud cover and cloud masking may also cause an inconsistency in the time interval of the obtained image. To ensure that the interval of time series for each pixel is consistent, the composite images for every 15 days were generated using the median value of all observations over the 15-day period (Oliphant et al., 2019; Wilson et al.,

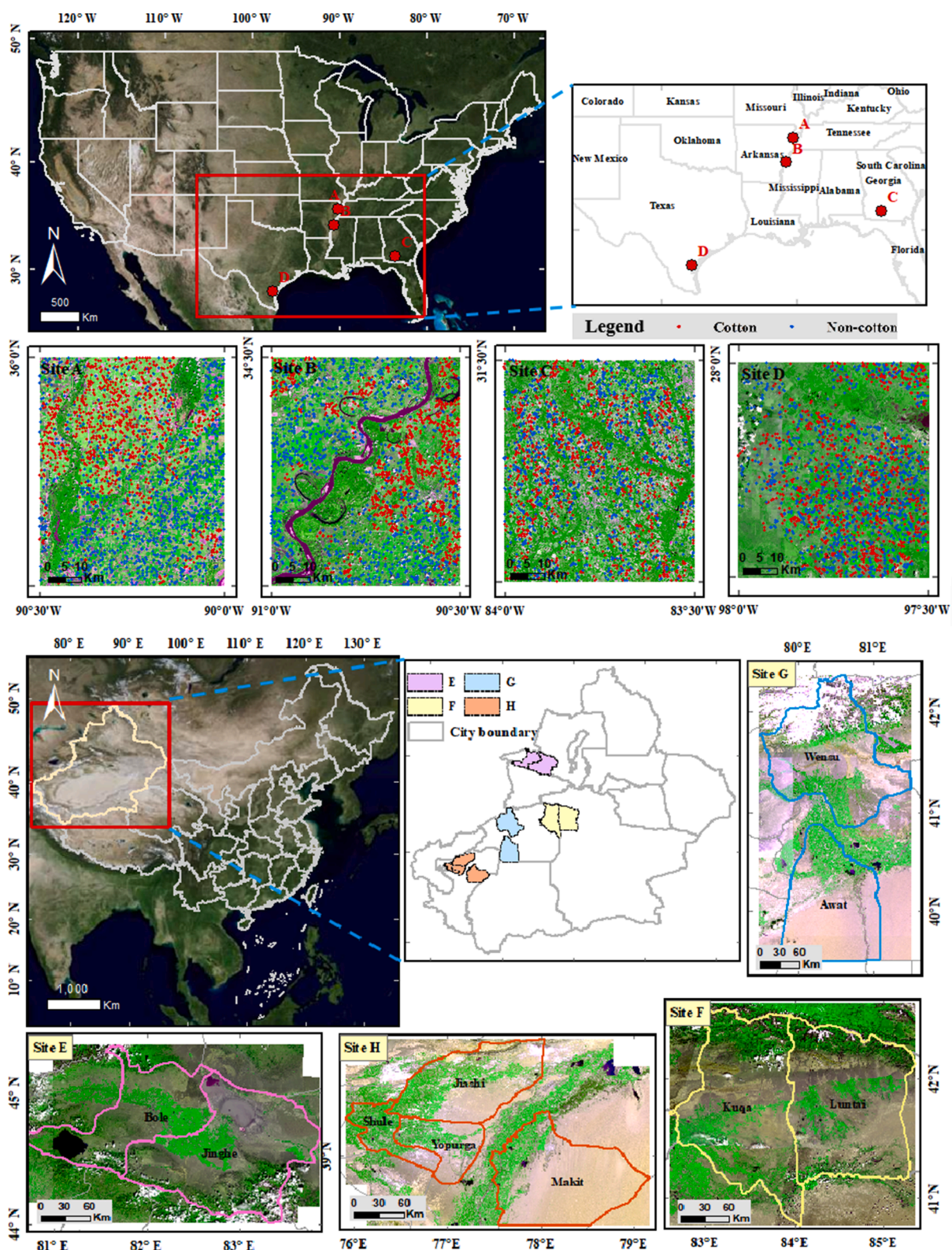
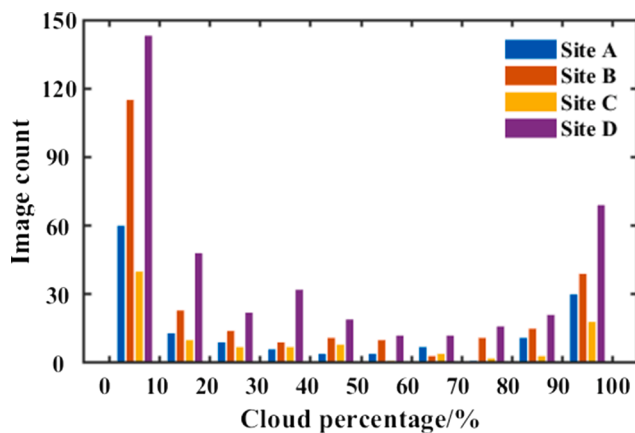


Fig. 2. Locations of study sites A-D and cotton/ non-cotton samples in the United States and sites E-H in Xinjiang of China as well as their RGB color composites of red, near infrared and blue. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

Main parameters of the spectral bands of Sentinel-2 used in this study.

Band No.	Band type	Central wavelength/nm	Spatial resolution/m	Abbreviation used in this study
B2	Blue	490	10	B
B3	Green	560	10	G
B4	Red	665	10	R
B5	Vegetation Red Edge 1	705	20	RE1
B6	Vegetation Red Edge 2	740	20	RE2
B7	Vegetation Red Edge 3	783	20	RE3
B8	Near infrared	842	10	NIR
B11	Shortwave infrared 1	1610	20	SW1
B12	Shortwave infrared 2	2190	20	SW2

**Fig. 3.** The number of images with different cloud percentages for sites A–D.

2018). Moreover, in some regions, high-quality observations may not be available because of cloud cover or other issues. The linear interpolation was used to fill these gaps using high-quality observation data before and after the time step.

Apart from the 9 spectral bands as above-mentioned in Table 1, the

18 vegetation indices (VIs), which were commonly used in literature on cotton studies, were also calculated and selected as the optical candidate features to develop the cotton identification rules in this study (Table 2).

2.2.2. Sentinel-1 images and preprocessing

The Sentinel-1A and Sentinel-1B satellites carry a dual-polarization SAR instrument at C-band, which can capture images with a spatial resolution of 5–40 m and a revisit period of 12 days. Four imaging modes, which include Strip Map (SM), Interferometric Wide-swath (IW), Extra Wide-swath (EW), and Wave (WV), are provided. The IW mode provides dual polarization imagery consisting of vertical transmit, vertical receive (VV) and vertical transmit, horizontal receive (VH). The Ground Range Detected (GRD) products acquired in IW mode were used in this study. The main parameters of the product are shown in Table 3. The images covering the study area during the study period were filtered on the GEE, which have been preprocessed to a calibrated, *ortho*-corrected product with a resolution of 10 m using the Sentinel-1 Toolbox (Jin et al., 2019). The Refined Lee Filter with a window size of 7×7 was applied to reduce noise in the Sentinel-1 images (Yomby et al., 2015). To keep consistent with Sentinel-2 data, the 15-day composited images were generated by calculating the median values of all observations over the 15-day period. In addition to the VV and VH polarizations, their combinations, such as VH/VV and VV – VH, were calculated and used as radar candidate features to distinguish cotton from other crops due to their potential for agricultural applications (Wang et al., 2020).

2.2.3. Reference samples

In this study, the Cropland Data Layer (CDL) from USDA was used as a reference map to select training and validation samples for the four study sites in the U.S. It is a raster, geo-referenced, crop-specific land cover data layer created annually for the continental United States using

Table 3

Main parameters of Sentinel-1 IW GRD product.

Parameter	Specification
Wavelength	C-band
Frequency	5.405 GHz
Spatial resolution	10 m
Repeat cycle	6 days (two satellites)
Number of looks (range $\times$ azimuth)	5×1
Ground range coverage	251.8 km
Swath width	250 km

Table 2

The vegetation indices used in this study and their expressions using Sentinel-2 bands.

Spectral index	Abbreviation	Expression using Sentinel-2 bands	Reference
Normalized Difference Vegetation Index	NDVI	$(B8 - B4)/(B8 + B4)$	(Huete et al., 2002)
Enhanced Vegetation Index	EVI	$2.5 \times (B8 - B4)/(B8 + 6 \times B4 - 7.5 \times B2 + 1)$	(Huete et al., 2002)
Infrared Percentage Vegetation Index	IPVI	$B8/(B8 + B4)$	(Crippen, 1990)
Transformed Normalized Difference Vegetation Index	TNDVI	$\sqrt{(B8 - B4)/(B8 + B4) + 0.5}$	(Tucker, 1979)
Soil Adjusted Vegetation Index	SAVI	$1.5 \times (B8 - B4)/(B8 + B4 + 0.5)$	(Huete, 1988)
Optimized Soil Adjusted Vegetation Index	OSAVI	$1.16 \times (B8 - B4)/(B8 + B4 + 0.16)$	(Rondeaux et al., 1996)
Green Atmospherically Resistant Vegetation Index	GARI	$(B8 - (B3 - (B2 - B4)))/(B8 + (B3 - (B2 - B4)))$	(Gitelson et al., 1996)
Green Normalized Difference Vegetation Index	GNDVI	$(B8 - B3)/(B8 + B3)$	(Gitelson and Merzlyak, 1998)
Sentinel-2 Red-Edge Position	S2REP	$705 + 35 \times (((B4 + B7)/2) - B5)/(B6 - B5)$	(Frampton et al., 2013)
Inverted Red-Edge Chlorophyll Index	IRECI	$(B7 - B4)/(B5/B6)$	(Frampton et al., 2013)
Red Edge Inflection Point	REIP	$700 + 40 \times (((B4 + B7)/2) - B5)/(B6 - B5)$	(Major, 1988)
MERIS Terrestrial Chlorophyll Index	MTCI	$(B6 - B5)/(B5 - B4)$	(Dash and Curran, 2007)
Pigment Specific Simple Ratio	PSSRa	$B7/B4$	(Blackburn, 1998)
Modified Chlorophyll Absorption Ratio Index	MCARI	$((B5 - B4) - 0.2 \times (B5 - B3))/(B5/B4)$	(Daughtry et al., 2000)
Normalized differential senescent vegetation index	NDSVI	$(B11 - B4)/(B11 + B4)$	(Yin et al., 2020)
Normalized difference residue index	NDRI	$(B4 - B12)/(B4 + B12)$	(Zhang et al., 2020)
Normalized difference tillage index	NDTI	$(B11 - B12)/(B11 + B12)$	(Yin et al., 2020)
Land surface water index	LSWI	$(B8 - B11)/(B8 + B11)$	(Yin et al., 2020)

moderate resolution satellite imagery and extensive agricultural ground truth (<https://nassgeodata.gmu.edu/CropScape/>). It includes crop type labels and is often used as a reference map for crop classification (Cai et al., 2018; Sun et al., 2019; Zhong et al., 2019). The spatial resolution of the CDL data during our study periods is 30 m. The CDL data for Arkansas, Mississippi and Georgia in 2019 as well as for Texas in 2020 have high self-reported accuracies for cotton class, with the user's and producer's accuracy higher than 84.1% and 89.1%, respectively. Moreover, the CDL data provide a confidence layer that reflects the per-pixel prediction confidence for a given classification, where 0 represents the least confident and 100 represents the most confident. In this study, the cotton and non-cotton samples in sites A-D were randomly selected from the region with the confidence value higher than 90%.

2.2.4. Statistical data

For Xinjiang, the county-level statistical cotton areas in 2018 were collected from the Xinjiang Statistical Yearbook, which were used to compare with the detected areas of the mapping results and to assess the performance of the proposed CMI.

2.2.5. Cropland mask

To rule out the interference of non-croplands areas (i.e., grass, forest, water, building bare land and others), the input satellite data were masked by the cropland layer. For the study sites of the U.S., a binary cropland mask was generated using the cultivated layer derived from CDL data and resampled from the original spatial resolution (30 m) to 10 m. For the study sites of Xinjiang, the cropland map with a spatial resolution of 10 m was extracted from the Finer Resolution Observation and Monitoring of Global Land Cover (FROM-GLC10) datasets (Gong et al., 2019).

2.3. Developing the CMI for cotton identification

Taking site A as an example, this section mainly describes the process of developing the CMI. The time series curves of spectral reflectance, vegetation index, spectral angle, and backscattering coefficients of cotton and other crops during the cotton growth period were preliminarily analyzed in Section 2.3.1–2.3.3, respectively. The three assumptions were then proposed in Section 2.3.4 to select the optimal features to construct the CMI.

2.3.1. Spectral reflectance of cotton and other crops

Phenology information is a significant characteristic of the crops, which can be exhibited in the time series of remotely sensed images and plays a valuable role in crop classification (Zhong et al., 2016). The mean spectral reflectance and NDVI curves for cotton, corn, rice, soybeans and others during April–November in site A are shown in Fig. 4, which are calculated from the 100 samples for each class. It can be seen that the differences between cotton and the other crops were not obvious in the visible spectral bands. The peak values of the RE2, RE3 and NIR of the cotton samples are higher than those of the other crops, which correspond to the time when the peak of NDVI occurs.

2.3.2. Spectral angle of cotton and other crops

In addition to spectral reflectance, the relationship between spectral bands, such as spectral angle, is also an important indicator in vegetation remote sensing research (Khanna et al., 2007). The angle index can not only capture the magnitude of spectral reflectance as the spectral index does, but also is related to the reflectance of adjacent bands (Palacios-Orueta et al., 2012). The spectral angle can reflect the relative relationship of the three bands and summarize the shape of the reflectance spectrum over the wavelength range of these bands. In the calculation of angles, the center of each spectral band is considered to be the vertex of a triangle, so the angle (α) at a certain vertex can be

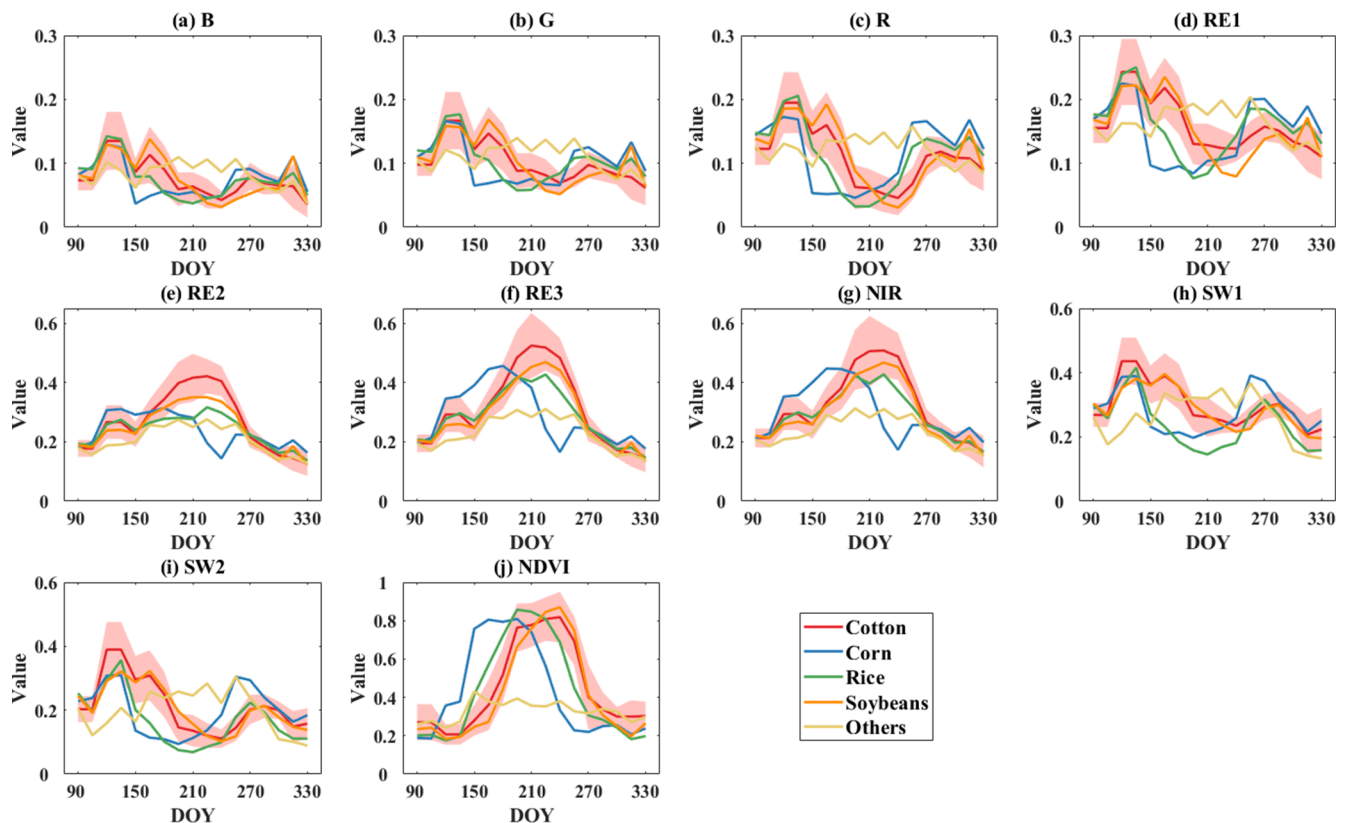


Fig. 4. Temporal profiles of spectral reflectance (a–i) and NDVI (j) of cotton, corn, rice, soybeans and others from April to November. The shading represents the standard deviation of each temporal profile for cotton samples.

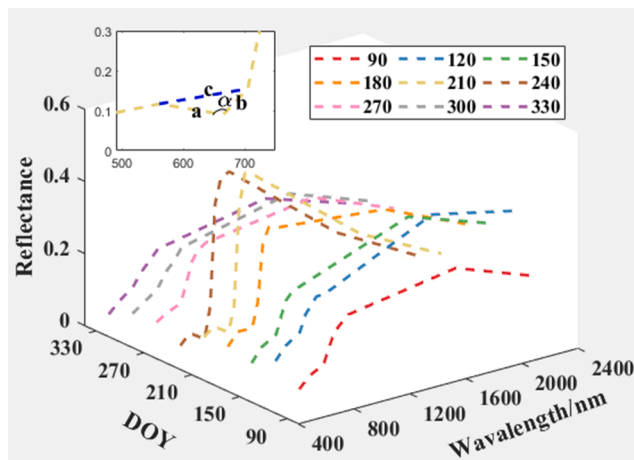


Fig. 5. Spectral curves of the cotton samples from April to November.

calculated using the law of cosines as follows (Khanna et al., 2007):

$$\alpha = \arccos \frac{a^2 + b^2 - c^2}{2ab} \quad (1)$$

where a , b and c represent Euclidian distances between three vertices. Taking the angle at red reflectance as an example, a , b and c correspond to the distances between the green and red, red and red-edge 1, as well as green and red-edge 1 band, respectively (Fig. 5).

The mean spectral curves of 100 cotton samples during the cotton

growth period are shown in Fig. 5. In the visible bands, the spectral reflectance of cotton showed a decreasing trend, while in the near infrared band, the spectral reflectance increased first and then decreased. The spectral angles at green, red, red-edge 1, red-edge 2, red-edge 3, near infrared, and shortwave infrared 1 bands for cotton and other crops during the cotton growth period are shown in Fig. 6. It can be seen that the spectral angles at green, red, and red-edge bands for the crops showed a trend of decreasing first and then increasing during the cotton growth period. During the period of minimum value for cotton, the spectral angles at red and red-edge 2 reflectance of cotton are much smaller than those of non-cotton, which correspond to the time when the peak of NDVI occurs.

2.3.3. Backscattering characteristics of cotton and other crops

The temporal profiles of the VV and VH polarizations as well as their combinations (i.e., VH/VV and VV-VH) for cotton and other crops are shown in Fig. 7, which are derived from the Sentinel-1 images. In general, the backscattering coefficients of dual-band cross-polarization (VH) were smaller than that of single co-polarization (VV). The backscattering coefficients in VV and VH for cotton were higher than those of other crops during the middle growth periods. The absolute values of backscattering coefficients in VH from June to August were more than twice that of VV of cotton.

2.3.4. Feature selection

The maximum greenness time is one of the key indicators in vegetation classification, which can be extracted from the time series of vegetation index (Pastor-Guzman et al., 2018). Based on the above

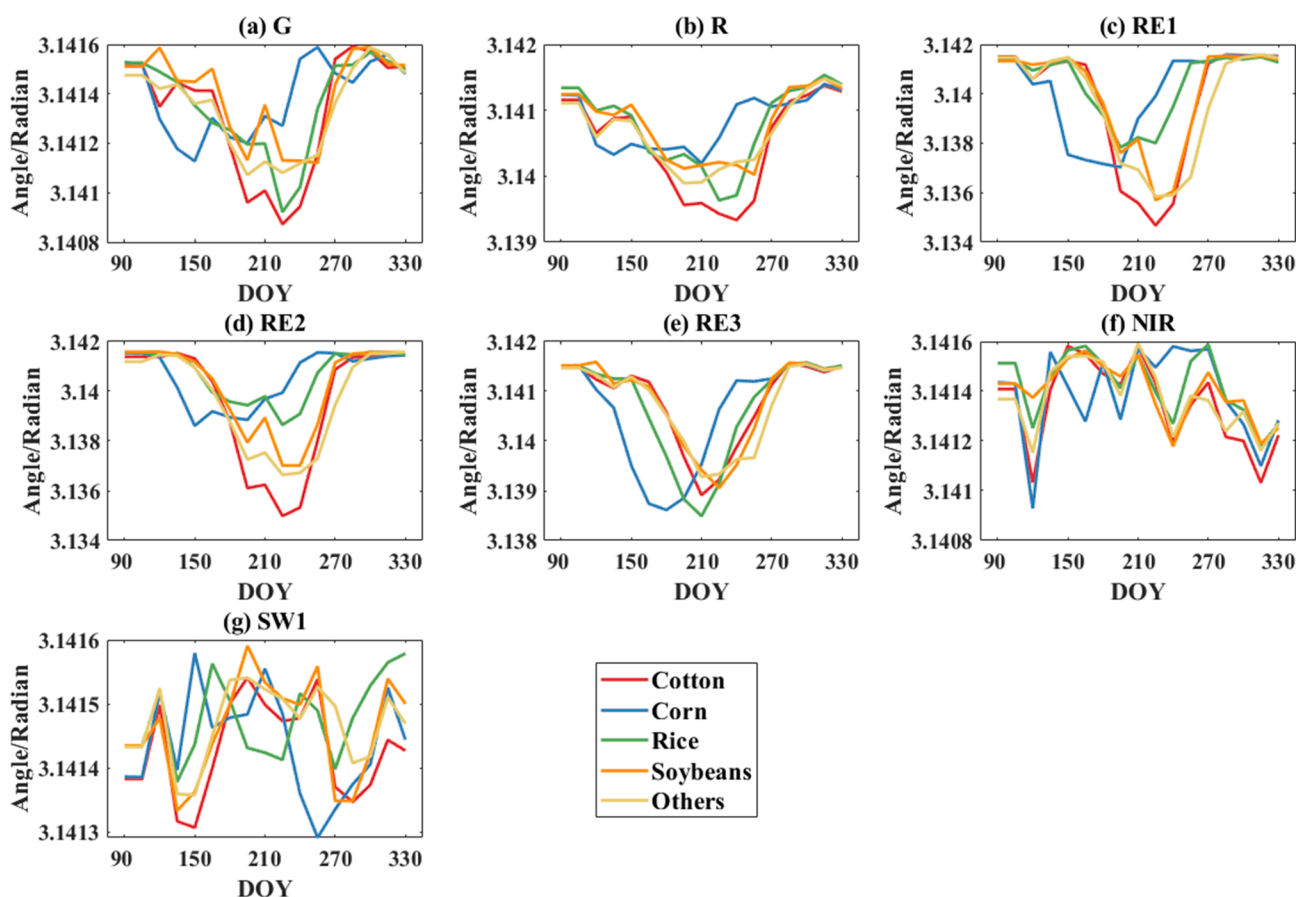


Fig. 6. Temporal profiles of spectral angles at green, red, red-edge 1, red-edge 2, red-edge 3, near infrared, and shortwave infrared 1 bands for cotton, corn, rice, soybeans and others from April to November. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

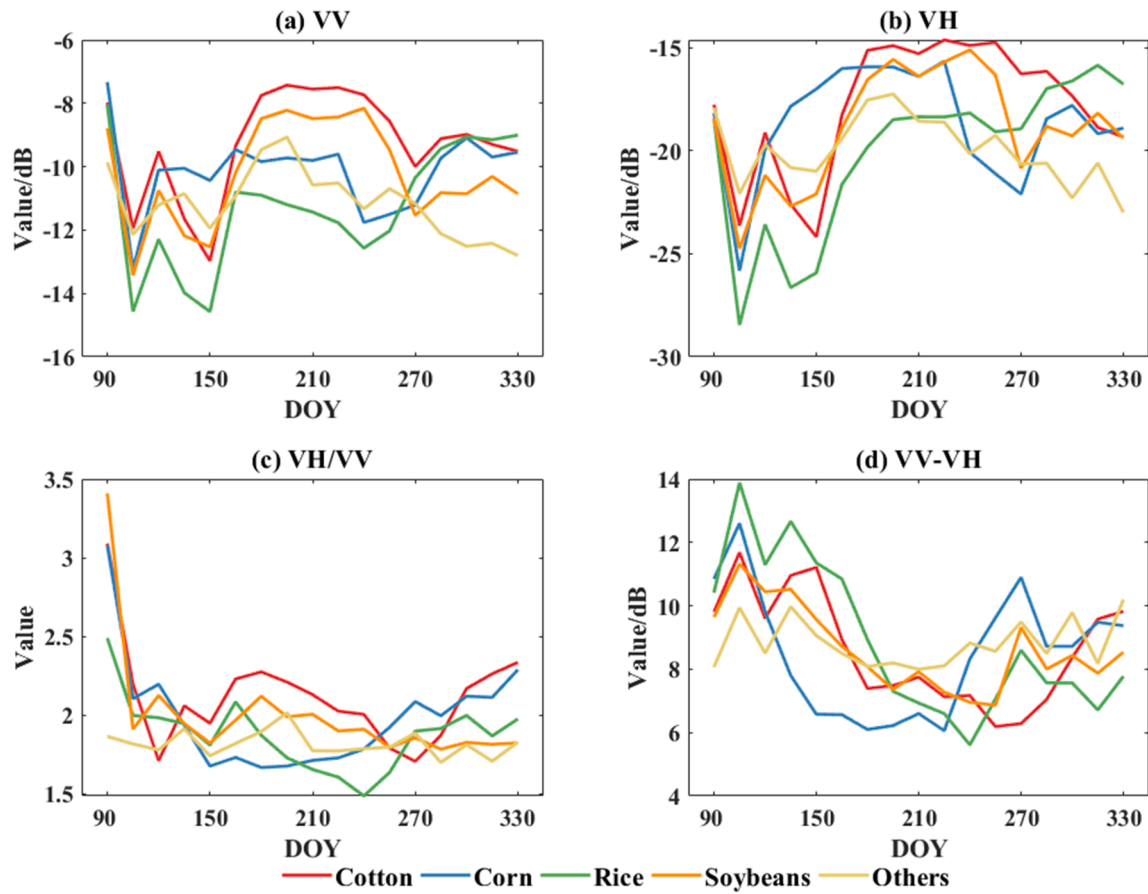


Fig. 7. Temporal profiles of VV, VH, VH/VV and VV-VH for cotton, corn, rice, soybeans and others from April to November.

preliminary analysis, the features at the peak greenness time (maximum NDVI) exhibit much difference between cotton and other crops, thus could be considered to be used to distinguish cotton from non-cotton crops. Moreover, considering the phenology differences of crops in different regions, an adaptive window was adopted to develop the CMI. Specifically, the features at the peak greenness time for each pixel were used to develop the CMI. To further determine the appropriate features to distinguish cotton from other crops, the following three assumptions were proposed.

Firstly, we assumed that cotton has a maximum or minimum value in a certain spectral band or vegetation index at the peak greenness time compared to other crops. In other words, cotton differs significantly from other crops in a certain feature, which is helpful to the identification of the cotton cultivated area. Thus, we proposed the following criteria to select feature 1 from the 27 optical candidate features mentioned in Section 2.2.1.

$$\text{feature } 1 = \arg\max_i(D_i) \quad (2)$$

where i represents the serial number of optical candidate features, ranging from 1 to 27. D_i represents the separability between cotton and non-cotton samples at the i th feature. In this study, two indicators including the M index (Kaufman and Remer, 1994) and Jeffries-Matusita (JM) distance (Swain and Davis, 1978) were used, which have been widely used in separability analysis (Guo and Li, 2020; Yang et al., 2017). The M index is defined as the absolute value of the difference of two mean values divided by the sum of the standard deviations of two classes, which can be calculated as follows (Guo and Li, 2020):

$$M = \frac{|\mu_1 - \mu_2|}{\sigma_1 + \sigma_2} \quad (3)$$

where μ_1 and μ_2 represent the mean values of two classes, σ_1 and σ_2 represent the standard deviations of two classes. For a certain feature, it indicates a good separability between the two classes when the M value is higher than 1, while the poorly separable when the M value is less than 1.

The JM distance between a pair of class-specific functions is given by:

$$JM = 2(1 - e^{-B}) \quad (4)$$

$$B = \frac{1}{8}(\mu_1 - \mu_2)^2 \frac{2}{\sigma_1^2 + \sigma_2^2} + \frac{1}{2} \ln\left(\frac{\sigma_1^2 + \sigma_2^2}{2\sigma_1\sigma_2}\right) \quad (5)$$

It indicates that the two classes are poorly separated when the JM value is less than 1 (Deng and Wu, 2012).

Similarly, the second and third assumptions were proposed that the spectral angle or radar feature was larger or smaller for cotton than that for other crops at the peak greenness time. Therefore, the feature 2 was selected from the 7 spectral angles including the angle at green, red, red-edge 1, red-edge 2, red-edge 3, near infrared and shortwave infrared 1. The feature 3 was selected from the 4 radar candidate features including VV, VH, VH/VV and VV - VH.

All the above proposed assumptions were used to select the optimal features to develop the CMI for cotton identification. Specifically, adequate number of samples of cotton (1000) and other crops (1000) in site A were randomly selected with reference to the CDL data. The peak greenness time for each sample was determined by the time when the

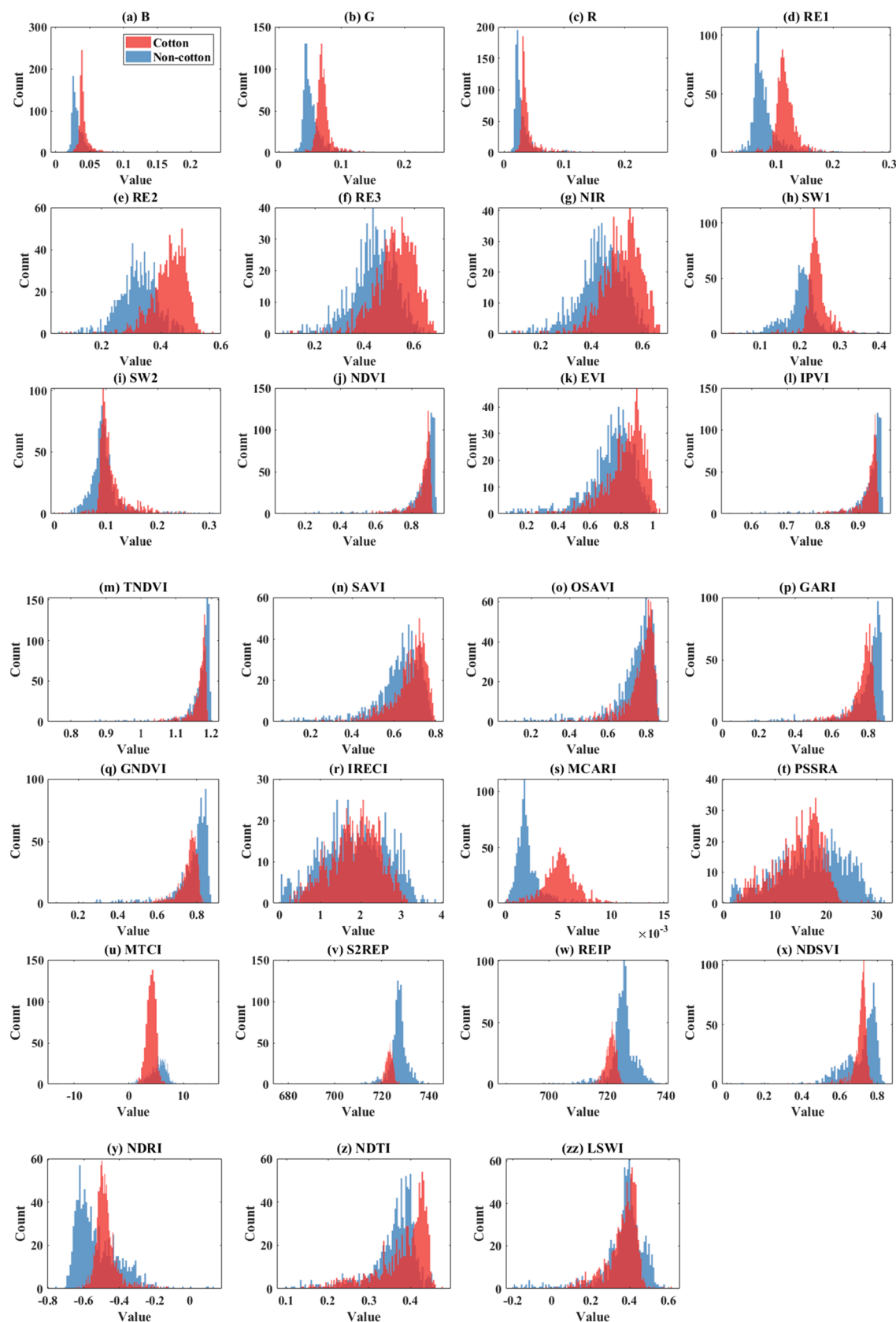


Fig. 8. Histograms of spectral reflectance (a-i) and vegetation index (j-zz) values of cotton and non-cotton samples at the peak greenness time.

maximum NDVI occurs. Then, the candidate feature values corresponding to the peak greenness time for each sample were obtained and analyzed to search for the optimal features that can discriminate cotton from other crops. The independent *t*-test was used to investigate the statistical significance of the differences between the mean values of optimal features for cotton and other crops. Finally, the CMI was constructed using these features.

2.4. Accuracy assessment and comparison

2.4.1. Accuracy assessment

In this study, for sites A–D, the classification accuracy of the mapping results achieved by the proposed CMI was assessed by comparing with the CDL map. The overall accuracy (OA), user's accuracy (UA) and producer's accuracy (PA) were calculated from the confusion matrix. Moreover, the F1-score was calculated using the UA and PA as follows (Guo and Li, 2020):

$$F1 - score = 2 \times \frac{UA \times PA}{UA + PA} \quad (6)$$

For sites E–H, the cotton areas detected by the CMI were compared with the statistical cotton areas at the county level. The mean relative error (MRE) and determination coefficient (R^2) between the detected and statistical cotton areas were calculated.

2.4.2. Accuracy comparison

To further assess the performance of the CMI, the accuracy of the cotton map obtained by the CMI was compared with those by the CEI (L. Wang et al., 2019) and three supervised classifiers (i.e., k-nearest neighbors (KNN), support vector machine (SVM) and random forest (RF)), respectively. The red-edge 1 and near infrared bands of the Sentinel-2 images were used to calculate the CEI.

As one of the simple machine learning algorithms, the principle of the KNN is that each sample can be represented by its nearest *K* neighboring values. The label of a sample is defined as the category corresponding to the majority of the *K* most similar samples in the feature space (Wilson et al., 2018). During the classification process, the category of a sample is only determined by the category of the nearest one or several samples (Chen et al., 2020). In this study, the *K* value was determined by comparing the classification accuracy under different *K* values. The results showed that the performance of the KNN in cotton mapping was optimal when the *K* was set as 4.

SVM is a non-parametric supervised classifier derived from statistical learning theory, which separates different classes by maximizing the

margin between them with a small number of training cases. It usually performs well in handling a limited number of training samples and processing high-dimensional data (Zhang et al., 2020). In this study, the radial basis function (RBF) kernel was used due to its' robust performance in remote sensing image classification (Hu et al., 2019). The kernel parameter γ and penalty factor *C* need to be set to train the SVM classifier. The former determines how the data is distributed in the new feature space and the latter controls the penalty associated with misclassified training samples. The parameters were set to the default values of the fitsvm function in Matlab software.

As an ensemble classifier, the RF adopts multiple decision trees and then assigns classification labels by majority vote. It has been widely used for land cover mapping and crop mapping based on remotely sensed images (Breiman, 2001; Pareeth et al., 2019; Zhang and Yang, 2020). Two parameters need to be set for the RF classifier, including the number of trees and the number of prediction variables used in each node to grow the tree. In this study, they were set to 500 and square root of the number of input variables, respectively (Belgiu and Drăgu, 2016). The time series NDVI during the cotton growth period was used as input into the KNN, SVM and RF classifiers, respectively. The training samples were randomly selected from the region with the confidence value of the CDL map larger than 90%.

3. Results

3.1. Development of the CMI

3.1.1. Analysis of the first assumption

To develop an efficient index to detect the cotton cultivated area, the objective of the first assumption was to select the optimal reflectance or vegetation index from the optical candidate features to discriminate cotton from other crops. The distributions of the 27 optical candidate features at the peak greenness time are shown in Fig. 8, which is based

Table 4

Independent *t*-test between cotton and other crops in terms of the mean values of feature 1, feature 2, CMI and CMI-S2 using 1000 samples for each class.

T-test between cotton and ~		Corn	Rice	Soybeans	Others
Number of samples		1000	1000	1000	1000
Sig (2-tailed)	feature 1	0.000	0.000	0.000	0.000
	feature 2	0.000	0.000	0.000	0.000
	CMI	0.000	0.000	0.000	0.000
	CMI-S2	0.000	0.000	0.000	0.000

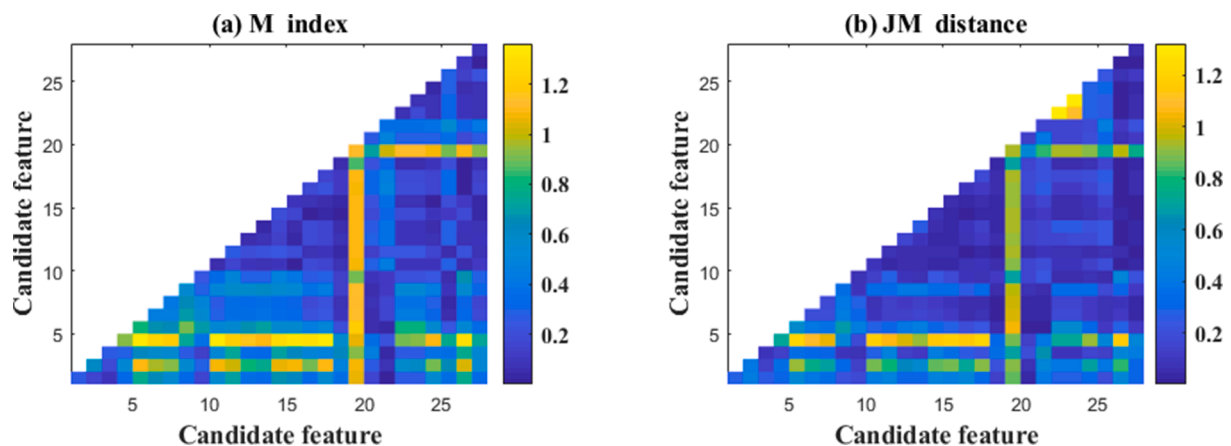


Fig. 9. The M index and JM distance between cotton and non-cotton samples using the multiplication of every two candidate features. The squares on the diagonal line represent the M and JM values using the single feature. The horizontal and vertical axes represent the 27 optical candidate features, i.e., B, G, R, RE1, RE2, RE3, NIR, SW1, SW2, NDVI, EVI, IPVI, TNDVI, SAVI, OSAVI, GARI, GNDVI, IRECI, MCARI, PSSRA, MTCI, S2REP, REIP, NDSVI, NDRI, NDTI, and LSWI.

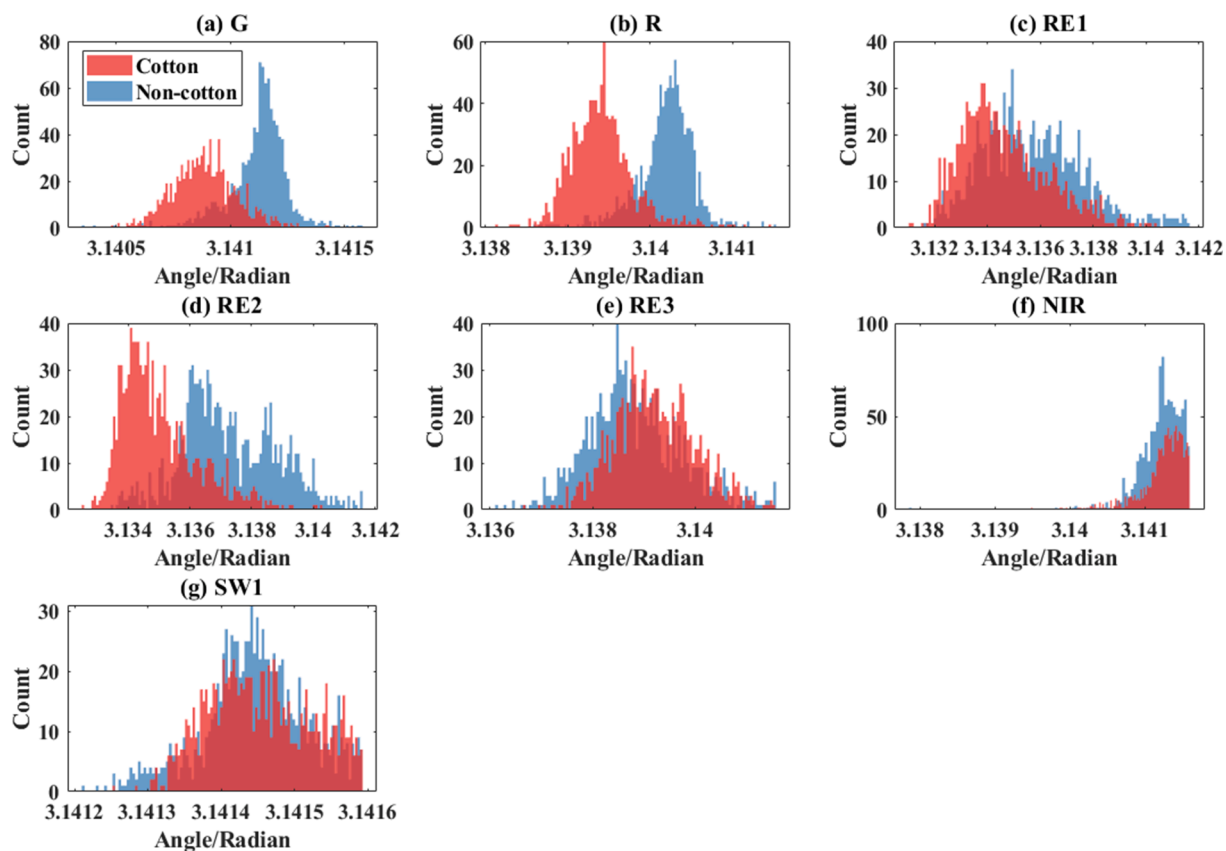


Fig. 10. Histograms of spectral angles of cotton and non-cotton samples at the peak greenness time.

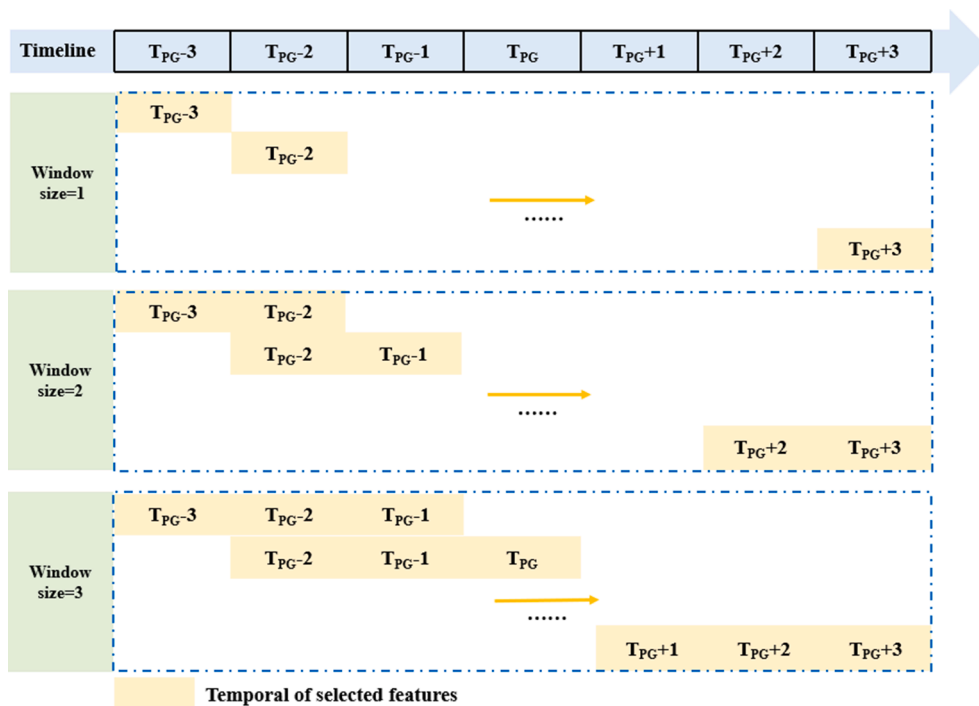


Fig. 11. The diagram of selection of feature 3 using the sliding window. T_{PG} represents the composite period of image when the maximum NDVI occurs.

on the 1000 cotton samples and 1000 non-cotton samples in site A. It can be seen that the differences between cotton and non-cotton samples for the RE1, RE2 and MCARI values are significant, with the M (JM) values of 0.87 (0.65), 0.86 (0.62) and 1.04 (0.85), respectively. The values of these features for most of the cotton samples were much larger than those for non-cotton samples. For the other optical candidate features, the overlap between the cotton and non-cotton samples was large, indicating the less informative for cotton identification.

Although the RE1, RE2 and MCARI performed better than other features in distinguishing cotton from non-cotton samples, the M and JM values were still much lower. Therefore, the addition and multiplication combinations of every two optical candidate features were tested to further amplify the differences between cotton and non-cotton samples. Due to relatively better results, the multiplication of two candidate features was adopted in this study to distinguish cotton from non-cotton. The M and JM values of the multiplication of every two features are shown in Fig. 9. It can be seen that the results incorporated the RE1 or MCARI were much better than those of others. Among all the combinations, the multiplication of RE1 and RE2 reflectance performed best, with the M and JM values of 1.36 and 1.20, respectively. Therefore, the multiplication of the red-edge 1 and red-edge 2 reflectance at the peak greenness time was selected as feature 1 to constitute the CMI for cotton identification. It is expected that this value will be maximized for cotton and minimal for other crops.

The independent *t*-test was employed to examine the statistical significance of the differences between mean values of feature 1 for cotton and other crops. As shown in Table 4, the sig (2-tailed) values are equal to zero, indicating that the differences between mean values of cotton and other crops are significant.

3.1.2. Analysis of the second assumption

The distributions of seven spectral angles of cotton and non-cotton samples at the peak greenness time were analyzed (Fig. 10). It can be seen that the differences of spectral angles at the red and green reflectance during the peak greenness time between cotton and non-cotton samples were significant, with the M (JM) values of 1.08 (0.88) and 0.88 (0.64), respectively. The values of these two features of cotton samples were much smaller than those of non-cotton samples. The spectral angle at the red reflectance was selected as feature 2 to construct the CMI due to its better separability. Moreover, the sig (2-tailed) values obtained by *t*-test were equal to zero, demonstrating that the differences between mean values of selected feature 2 for cotton and other crops were significant (Table 4).

3.1.3. Analysis of the third assumption

Due to the relatively low separability when using radar features alone, we investigated the performance of combining radar features with feature 1 and feature 2 for cotton mapping (Section 3.1.4). A sliding window was adopted to search for the optimal feature 3 that could perform best when combined with feature 1 and feature 2. Specifically, the features within different temporal windows derived from Sentinel-1 were tested. As shown in Fig. 11, the window size was tested from 1 to 3, with the window sliding around the maximum NDVI period. According to the experiments, the summation of backscattering coefficients in VV polarization during the peak greenness and previous two composite periods was selected as the feature 3.

3.1.4. Construction of CMI

According to the above results, for cotton, the multiplication of the reflectivity of red-edge 1 and red-edge 2 was higher than that of the other crops at the peak greenness time, while the spectral angle at red reflectance was lower than that of the other crops. The absolute value of summation of the backscattering coefficients in VV polarization was lower for cotton than that for other crops. To further amplify the differences, the larger value was used as the numerator and the smaller value as the denominator to constitute the Cotton Mapping Index (CMI)

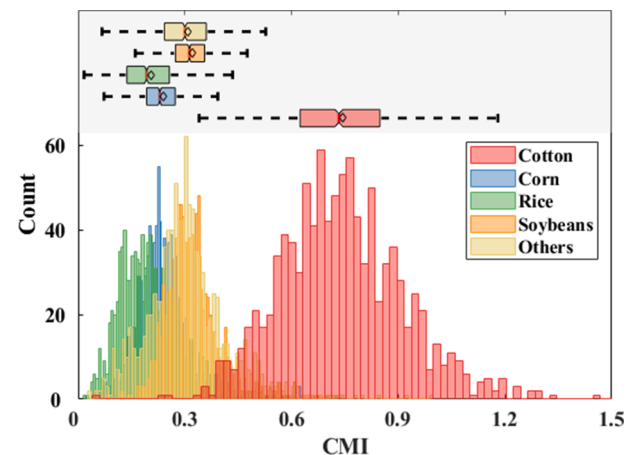


Fig. 12. The distribution of CMI values of cotton and non-cotton samples in site A. For each box, the central mark and diamond indicate the median and mean value, respectively. The left and right edges of the box represent the 25th and 75th percentiles, respectively.

for separation of cotton from other crops as follows:

$$CMI = \frac{\rho_{RE1}^{PG} \times \rho_{RE2}^{PG} \times \theta}{\alpha_R^{PG} \times \text{abs}(\sum_{t=PG-2}^{PG} \beta_{VV}^t)} \quad (7)$$

where ρ and α represent the spectral reflectance and spectral angle, respectively. β represents the backscattering coefficients. θ is a constant variable, and is set to 1000 to avoid too many decimals of the CMI in this study. t represents the composite period of the image, and abs represents absolute value.

Moreover, to test the performance of the combination of radar features and optical features in cotton mapping, we also constructed an index using spectral features alone for comparison, as follows:

$$CMI-S2 = (\rho_{RE1}^{PG} \times \rho_{RE2}^{PG}) / \alpha_R^{PG} \quad (8)$$

It is expected that the CMI and CMI-S2 values will be maximized for cotton and minimal for other crops.

To investigate the differences in CMI values between cotton and non-cotton crops, the 1000 samples for cotton, corn, rice, soybean and others were randomly selected from the CDL map with the confidence value higher than 90% in site A. The distribution of the CMI values for these samples is shown in Fig. 12. It can be seen that the mean CMI values of cotton are obviously larger than those of other crops. The mean values of CMI for cotton, corn, rice and soybeans were 0.74, 0.24, 0.21 and 0.32, respectively. The M and JM values between cotton and non-cotton samples were 1.45 and 1.30, respectively, indicating the relatively good separability between the cotton and non-cotton samples. Moreover, the sig (2-tailed) values obtained by *t*-test were equal to zero, demonstrating that the differences between mean values of CMI for cotton and other crops were significant (Table 4).

3.2. Determination of the optimum threshold value

An empirical method was adopted to select the appropriate threshold value of the CMI, which has been successfully used to determine the threshold in previous studies (Ashourloo et al., 2019). A total number of 2000 samples were randomly selected from the CDL map with the confidence value higher than 90% in site A, including 1000 cotton samples and 1000 non-cotton samples. The classification accuracy corresponding to different CMI thresholds with step size of 0.02 were calculated. The experiment was repeated 20 times. As shown in Fig. 13 (a), with the increase of the threshold value of CMI, the user's accuracy, overall accuracy and F1-score showed a trend of increasing first and then decreasing. When the threshold value was small, the omission error for

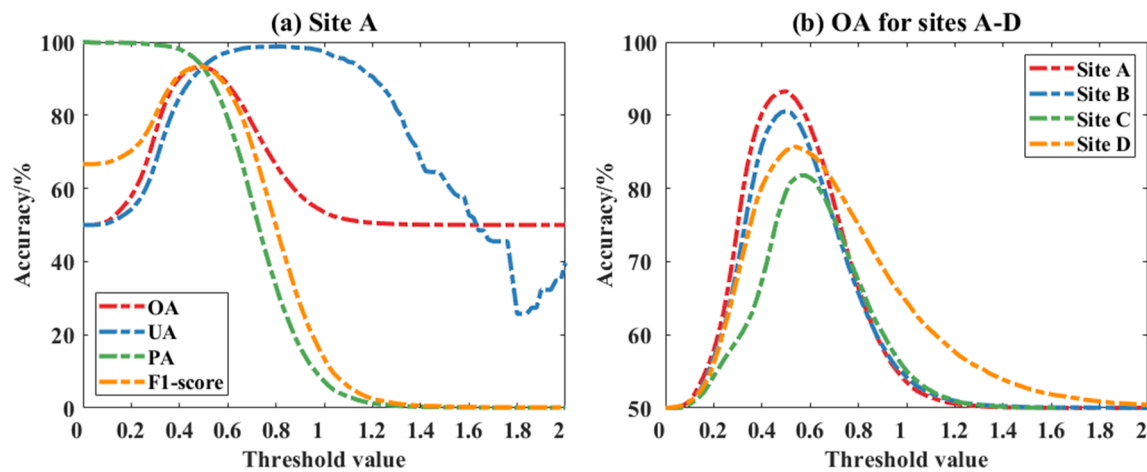


Fig. 13. Variations of classification accuracy with different CMI thresholds for sites A-D.

Table 5

Confusion matrix for classification results with the CMI threshold value of 0.50 for study sites A-D.

Study sites	CDL map	CMI map		PA (%)	UA (%)	OA (%)	F1-score (%)
		Cotton	Non-cotton				
A	Cotton	935	65	93.50	93.03	93.25	93.27
	Non-cotton	70	930				
B	Cotton	886	114	88.60	90.69	89.75	89.63
	Non-cotton	91	909				
C	Cotton	873	127	87.30	77.81	81.20	82.28
	Non-cotton	249	751				
D	Cotton	878	122	87.80	86.33	86.95	87.06
	Non-cotton	139	861				

cotton was small due to the relatively large CMI values of cotton, thus the producer's accuracy was relatively high. With the increase of threshold value, the cotton samples with low CMI values could not be identified, thus the producer's accuracy showed the decreasing tendency. In this study, the overall accuracy reached the peak in site A when the CMI threshold was 0.5. Thus, it was considered to be an appropriate threshold and used for the identification of cotton cultivated areas in other sites. The same method was applied to CMI-S2, and the optimal threshold value for CMI-S2 was 0.013 in site A.

3.3. Accuracy assessment of the proposed CMI

3.3.1. Accuracy assessment for sites A-D

As shown in Table 5, when the threshold value of CMI was set as 0.50, the overall accuracy and F1-score of the CMI-derived cotton map for site A were 93.25% and 93.27%, respectively. Then it was applied to other sites to further verify the validity of the CMI and the threshold. The results showed that the overall accuracies for sites B, C and D were 89.75%, 81.20% and 86.95%, respectively. The producer's accuracy was higher than the user's accuracy for sites A, C and D, indicating that the omission error was lower than the commission error in these sites. The producer's accuracy was lower than the user's accuracy for sites B, indicating that the commission error was lower than the omission error in this site.

In this study, the samples with confidence values higher than 90% were randomly selected from the CDL map to develop the CMI and assess the accuracy for sites A-D (Table 5). In addition, the consistency between the CMI-derived cotton map and the CDL map at different confidence

intervals was also analyzed. As shown in Fig. 14, the consistency increases with the increase of the confidence values of the CDL map, indicating that the mapping results derived by the CMI have good accuracy.

The CMI-derived cotton maps as well as the consistency maps between the CDL and CMI-derived maps for sites A-D are shown in Fig. 15. In general, the distributions of cotton cultivated areas on CDL and CMI map were similar for four sites. As shown in Fig. 15 (e-h), the red and green areas represent the consistent cotton and non-cotton areas between the CDL and CMI map, respectively, representing the regions where CMI has a relatively high accuracy. Compared with the CDL map, the cotton area detected by the CMI was overestimated, and most of these cotton areas corresponded to the uncertain regions with confidence value less than 90% on the CDL map.

3.3.2. Application of CMI in Xinjiang sites

The four sites in Xinjiang were used to further verify the applicability of CMI for cotton mapping. The cotton areas were identified using the optimal threshold value of the CMI obtained in the site A. The MRE and the R^2 between the statistical and CMI-detected cotton areas were 43.44% and 0.74 at the county level, respectively. As shown in Fig. 16, most of the points are above the 1:1 line, indicating that the CMI slightly overestimates the cotton area when the threshold value is 0.50. Therefore, the threshold value of the CMI was fine-tuning using an empirical method. The R^2 and MRE achieved by different threshold values were calculated, with the step of 0.02 from 0.5 to 0.7. The results showed that the optimal threshold value for these regions was 0.56, with the R^2 and MRE of 0.60 and 26.69%, respectively. Thus, this threshold value was adopted for cotton mapping in these regions in 2018 (Fig. 17).

3.4. Comparison with other methods

The performance of the CMI was further assessed by comparing the classification accuracy of the CMI-derived map with those achieved by the other two indices (CMI-S2, CEI) and three supervised classifiers (KNN, SVM, RF) using the validation samples at sites A-D. The 20 groups of validation samples were randomly generated, and each group consisted of 1000 cotton and 1000 non-cotton samples. Considering the differences of phenology in different regions, the optimal CEI threshold and optimal temporal for each region were determined by empirical method.

As shown in Fig. 18, the CMI performed better than the RF, SVM and KNN classifiers in terms of the mean values of overall accuracy and F1-score in site A and site B, indicating that the performance of CMI without using training samples was similar to those of supervised classifiers. The overall accuracy and F1-score achieved by the CMI and CMI-S2 was

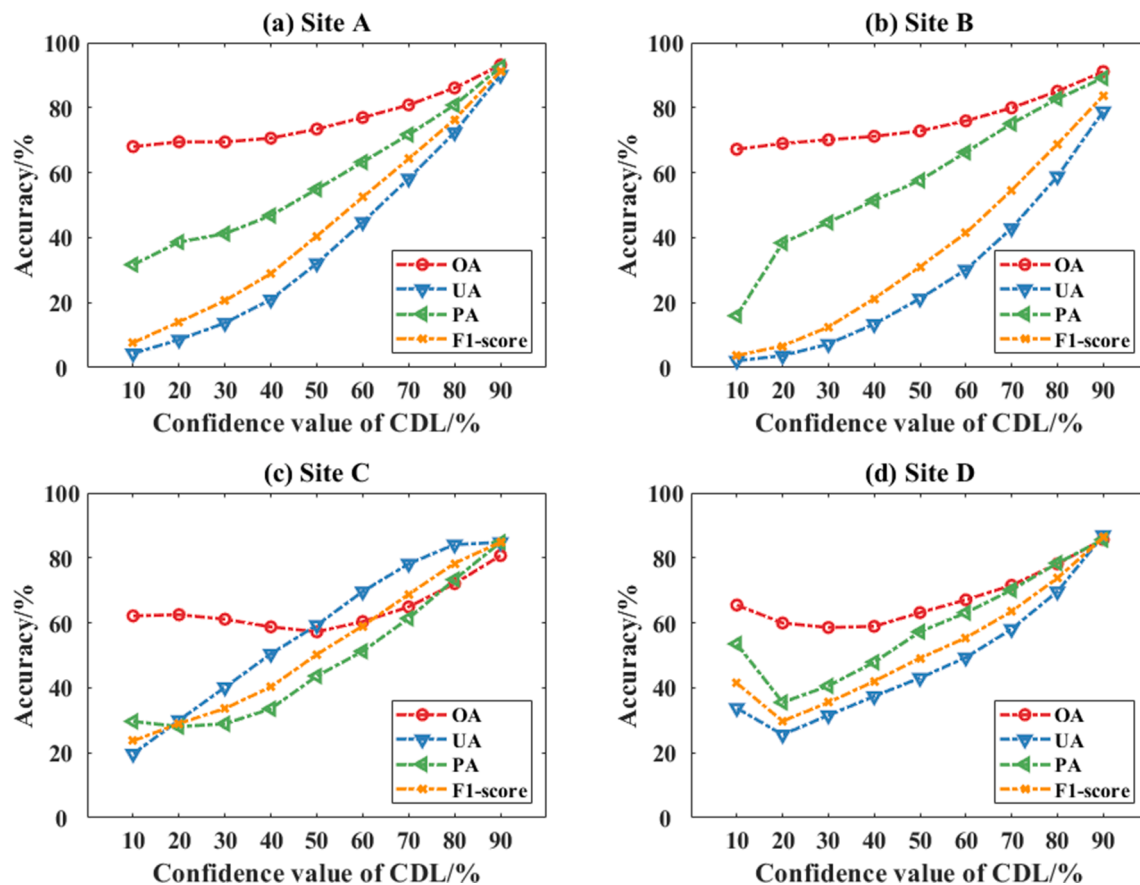


Fig. 14. Consistency between the CMI-derived and CDL-derived cotton map at different confidence intervals of the CDL map.

similar, both of which were superior to that of the CEI. Also, both the optimal CMI and CMI-S2 thresholds were determined in site A and then applied for four sites, while the optimal CEI threshold with the optimal temporal of image was searched first and then used for each site. These results indicated the advantages of adaptive window of the proposed CMI and CMI-S2. In site C and site D, the overall accuracy and F1-score achieved by three indices were lower than those of the KNN, SVM and RF classifiers, indicating that the supervised classifiers were robust when the training samples were available. In the absence of samples, among the three indices, the performance of CMI was significantly higher than that of CMI-S2 and CEI, indicating the advantages of the combination of optical and radar features in cotton mapping.

4. Discussion

4.1. Advantages of the proposed index

As an important cash crop and strategic material in the world, the spatial distribution information of cotton is of great significance to the monitoring of cotton growth and yield estimation. In recent years, machine learning algorithms have been proved to be an effective method to map the spatial distribution of various crops (He et al., 2019). However, most of these methods belong to supervised classifiers, which strongly depend on the samples to train the classifier (Dong et al., 2020). It still remains a challenge in cotton mapping when the samples are unavailable. There is a great demand for developing a method that can be used for automated mapping the cotton cultivated area.

To this end, a novel index termed CMI, was proposed using the time series of Sentinel-1 SAR and Sentinel-2 MSI images, and its ability for cotton mapping was examined in the eight sites selected from the U.S. and China. The reflectance of red-edge 1 and red-edge 2 bands, spectral

angle at the red band as well as backscattering coefficients in VV polarization were used to develop the CMI. The results demonstrated that the CMI can be effectively used to map the cotton distribution in the eight sites of this study without the use of training samples. Specifically, the CMI has the following advantages over existing methods.

Firstly, in this study, the time series of Sentinel-1 and Sentinel-2 images during the cotton growth periods were used as the initial datasets. The features at/around the peak greenness time for each pixel were used to develop the CMI for cotton identification, which was based on an adaptive window. The phenology differences in different regions can be considered in this study when compared with that using the single temporal image solely. The comparison results between the CMI-S2 and CEI in Fig. 18 demonstrated this advantage.

Secondly, the features derived from both Sentinel-2 MSI and Sentinel-1 SAR images were jointly adopted to develop the CMI and contribute to the ability to identify the cotton cultivated area. Previous studies have shown the potential of the red-edge band in crop classification using supervised classifiers (Zhang et al., 2020). The analysis in this study showed that the multiplication of red-edge 1 and red-edge 2 reflectance for cotton samples were much larger than those for non-cotton samples at the peak greenness time. Therefore, these two bands were chosen to develop the CMI based on the first assumption of this study. In addition to the spectral reflectance, the spectral angle is also an important indicator in vegetation studies (Khanna et al., 2007; Palacios-Orueta et al., 2012). Based on the second assumption of this study, the analysis results showed that the spectral angle at the red band of Sentinel-2 data for cotton samples were much smaller than those for non-cotton samples at the peak greenness time. With the above two groups of optical features, the optimal radar features were explored. The results showed that the incorporation of VV performed better than others for cotton identification in this study. The combination of optical

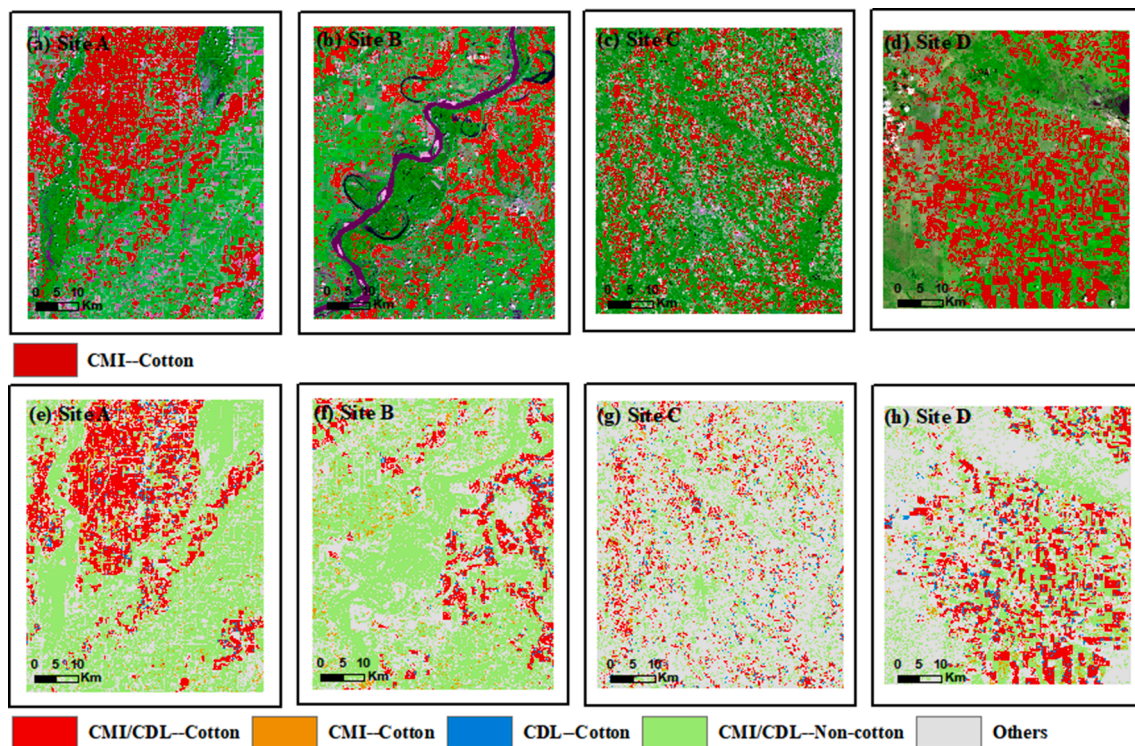


Fig. 15. False color composites of red, near infrared and blue of Sentinel-2 images overlaid with cotton maps detected by the CMI (in red color) for sites A-D (a-d). (e)-(h) show the comparison of the CMI and CDL map. 'Others' represents the regions with confidence value less than 90% of the CDL map. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

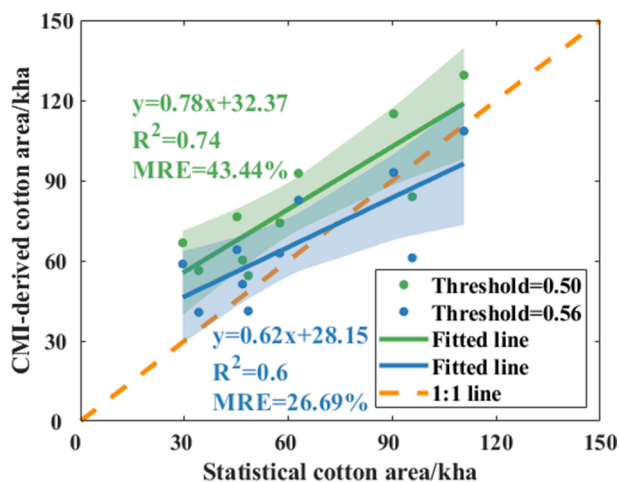


Fig. 16. Comparison of CMI-derived and statistical cotton area at the county level with the CMI thresholds of 0.50 and 0.56, respectively.

and radar features performed better than using optical features solely in cotton identification. The comparison results between the CMI and CMI-S2 in Fig. 18 demonstrated this advantage.

Third, compared to the supervised classifiers, the proposed CMI is convenient and performs well even if without training samples in identifying the cotton cultivated area, which is a significant advantage given the lack of survey samples available. Taking site A and site B as an example, the classification accuracies of the KNN, SVM and RF classifiers under different number of training samples were also compared. The 20 groups of training samples were randomly generated to train the classifier, and the classification accuracy was evaluated with the same validation samples. As shown in Fig. 19, with the increase of the number of training samples, the mean values of overall accuracy show an

increasing trend and gradually stabilized. The variation range of overall accuracy reduces with the increase of the number of training samples, indicating that there is still a high uncertainty in the case of a small number of training samples for supervised classifiers.

Moreover, timeline is an important factor when considering a crop type map, as the early map before the end of the season can help monitor growth, determine crop drought conditions, manage irrigation and predict crop yields, thus benefiting farmers and policymakers (Inglada et al., 2016; Vaudour et al., 2015). Taking sites A-D as an example, the satellite images with different time series lengths were used to test the potential of CMI in early season cotton mapping. The input data started in April and increased by half a month at a time. The classification accuracy achieved by input data with different lengths were compared (Fig. 20). With the length of time series increases, the overall accuracy showed a trend of increase. For sites A-C, the accuracy reached the maximum and gradually stabilized after August, indicating that using satellite data during day of year (DOY) 90–240 could achieve an acceptable accuracy in these sites. For site D, the accuracy tends to be stable using the time series data during DOY 90–180, which was earlier than for sites A-C. The reason is that the peak greenness time of cotton in site D (the southern Texas State) is much earlier than those in the other three sites. This indicates that the cotton map can be obtained before harvest using the CMI.

4.2. Uncertainty and potential refinements

Although with the above-mentioned advantages of the proposed method, there are still some limitations and uncertainties in cotton mapping. Firstly, in this study, the threshold value of the CMI was determined as 0.50 at site A, and its performance at sites B-D was acceptable. This indicated that this threshold can be adopted in the region where the samples are not available to determine an appropriate threshold value. However, as shown in Table 5, the accuracy of the mapping results in sites B-D using this value was slightly lower than that

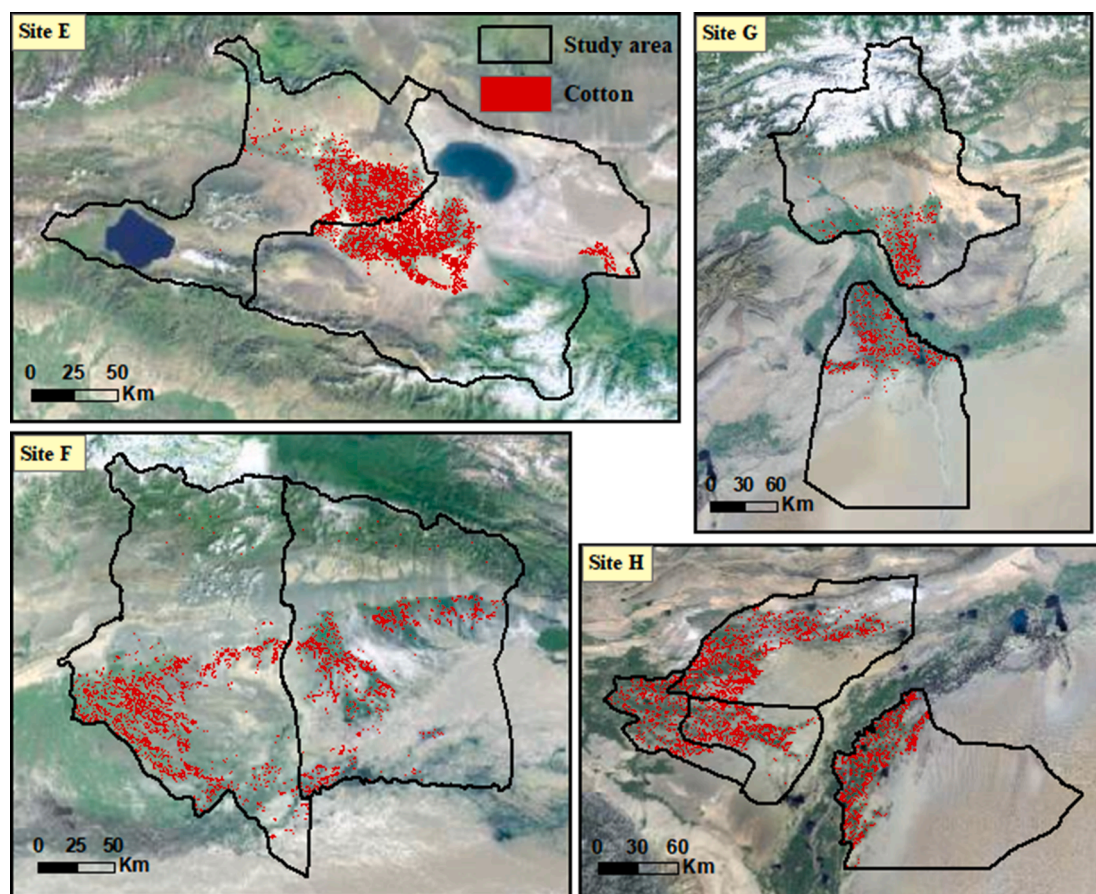


Fig. 17. The cotton distribution map detected by the CMI overlaid with the Google images for sites E-H.

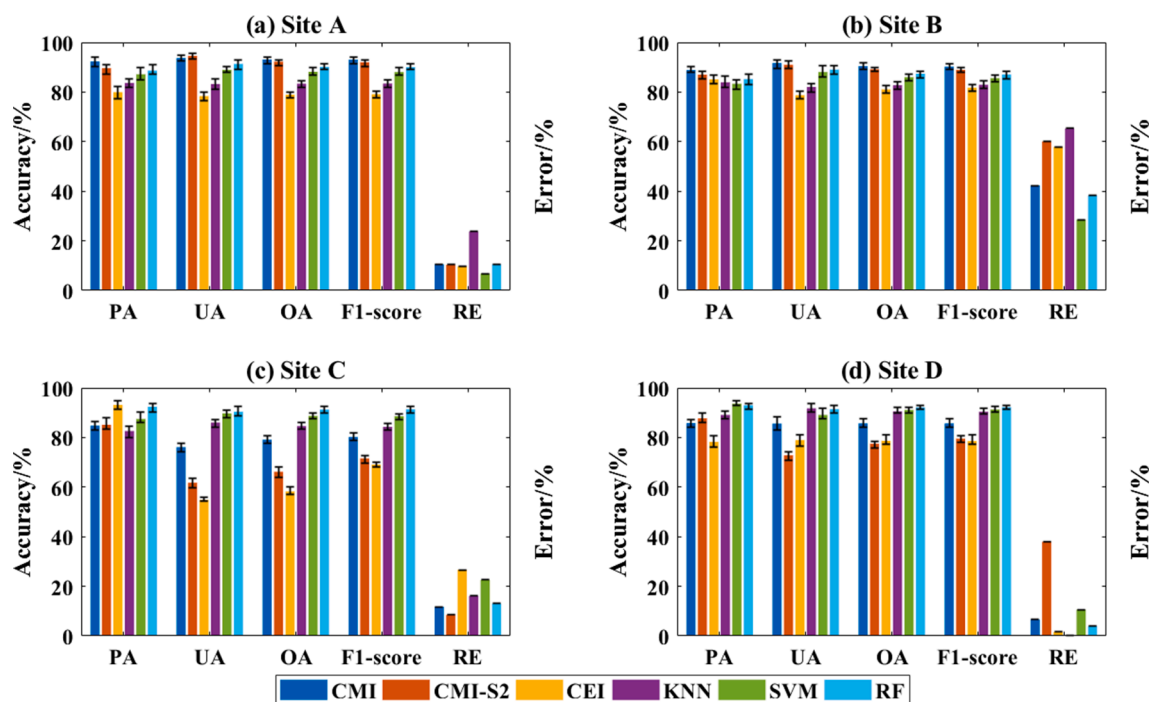


Fig. 18. Classification accuracy of cotton mapping results achieved by the CMI, CMI-S2, CEI, KNN, SVM and RF methods in sites A-D, respectively. RE represents the absolute value of relative error of cotton pixels between the mapping results and CDL map, which corresponds to the right vertical axis. The error bar represents the upper and lower bound of the accuracy assessed by 20 groups of validation samples.

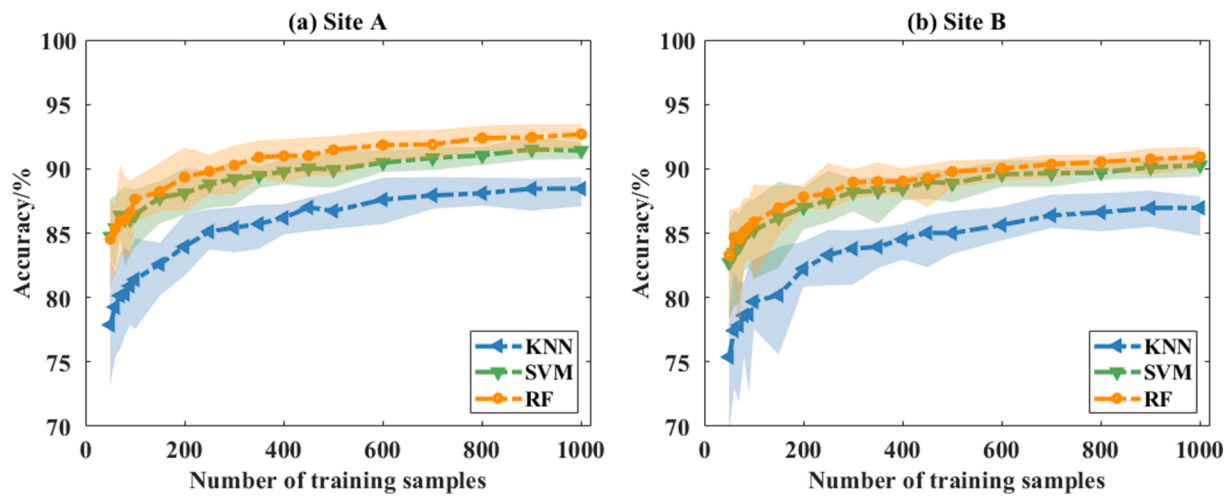


Fig. 19. Overall accuracy achieved by the KNN, SVM and RF classifiers with different numbers of training samples.

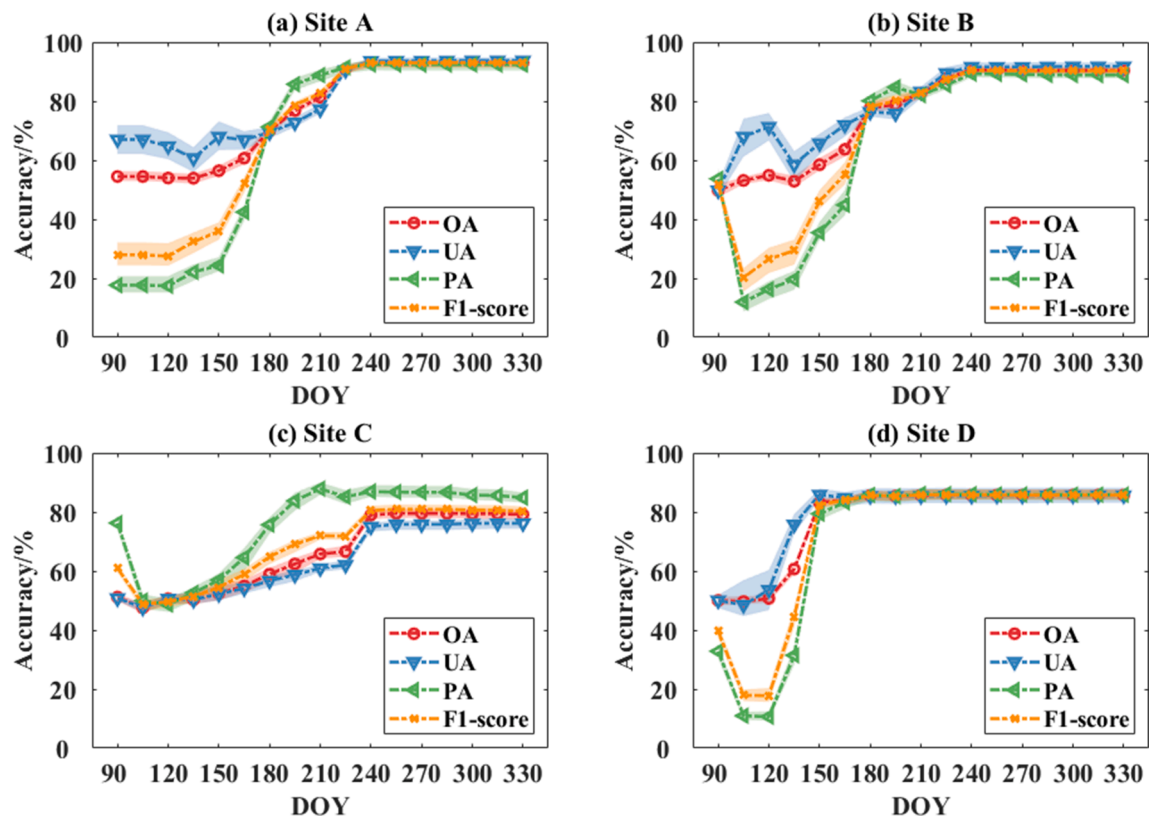


Fig. 20. Variations in the classification accuracy of the CMI-derived results obtained by using satellite images with different time series lengths in sites A-D.

in site A. Due to the availability of the adequate samples for cotton and other crops, the optimal CMI thresholds for sites B-D were also determined using the same method in Section 3.2 (Fig. 13 (b)). The results showed that the optimal threshold value for site B was the same as that for site A. When the threshold value was 0.58 and 0.54, the overall accuracy reached the peak for site C and site D, respectively, both of which were much higher than that using the threshold value of 0.50. For sites E-H, the evaluation results based on the county-level statistics showed that the R^2 was high whereas the cotton area was a little overestimated by the CMI with the threshold value of 0.50. It achieved a better performance using the threshold value of 0.56, which was fine-tuned using statistical data. The reason may be that the performance of the CMI was

slightly affected by different crop types at different sites. The results indicated that better performance may be achieved when a small number of reference data (i.e., training samples or statistics) can be used to fine-tune the CMI threshold. The applicability of the CMI and the determination of threshold in other regions need to be further assessed in the future study. More features could be explored and selected to further improve the performance of the CMI as well as the stability of the optimal CMI threshold for automatic cotton mapping.

Secondly, in the current study, the development and application of the CMI were only considered in the single cropping systems. According to the CDL map, there is a small percentage of double cropping system area in the sites A-D. Taking site A as an example, the areas of winter

wheat-soybeans, winter wheat-corn and winter wheat-cotton account for 0.488%, 0.019% and 0.007% of the whole study area, respectively. Among the pixels of these three classes, 96.80% were correctly classified using the CMI. Considering the relatively small percentage of double cropping system area in the current study sites, the applicability of the CMI in double/multiple cropping systems needs to be further explored in the future study. It is important for the application of the CMI in the cotton producing countries such as China (Yellow River basin and Yangtze River basin), India, Pakistan and Brazil with much double/multiple cropping systems areas. Moreover, the red-edge 1 and red-edge 2 bands of Sentinel-2 were used to calculate the CMI, both of which were resampled from 20 m to 10 m. The uncertainty of the resampling was not considered in this study. In the future research, the influence of resampling on classification accuracy can be further analyzed in cotton identification.

5. Conclusion

In this study, a novel index termed Cotton Mapping Index (CMI) was developed using Sentinel-1 SAR and Sentinel-2 MSI images, which was a convenient method for automatic identification of cotton cultivated area without training samples. The characteristics of temporal profiles of cotton samples during its growth period were analyzed to develop the CMI. Based on the analysis results, the reflectance at the red-edge 1 and red-edge 2 bands, the spectral angle at the red band, as well as back-scattering coefficients in VV polarization, were combined to constitute the CMI. Four sites from the U.S. and four sites from China were used to assess the performance of the CMI. The results indicated that the cotton cultivated area can be identified using the CMI with the overall accuracy higher than 81.20% for the sites from the U.S. as well as the mean relative error of 26.69% for the sites from Xinjiang of China. Compared with the supervised classifiers such as KNN, SVM and RF, the proposed CMI achieved a similar performance without using the training dataset. Compared with the CMI-S2 and CEI using optical features only, the CMI, which incorporated the optical and radar features, achieved a better performance in cotton mapping. The comparison results of the CMI, CMI-S2 and CEI also demonstrated the advantage of using the features within an adaptive window for cotton mapping. Moreover, the cotton cultivated area can be identified before harvest using the CMI. With the promising results of this study, the applicability of the CMI in other regions with different cropping systems and crop types needs to be further assessed in the future study.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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